

Prashant

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EDUCATION

- Vellore Institute of Technology**
 - Bachelor of Technology - Computer Science and Engineering; CGPA: 8.69*
 - Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Databases, Computer Networks*
- Bhopal, India
Sep 2023 - May 2027

SKILLS

- Programming:** C++ , OOPS , JavaScript , Problem Solving
- Backend:** Node.js , Express.js , MongoDB , REST APIs , Authentication & Authorization
- Data Structures and Concepts:** Data Structures, Algorithms, Object Oriented Programming, Database Management System, Operating System, Computer Networks
- Databases:** MySQL / SQL, Basic query optimization, joins
- Modeling:** UML, ER Diagrams, Data Flow Diagrams
- DevOps & Tools:** Git, AWS Deployment(Decent), Postman, MS-SQL, Firebase,Excel,Render,Vercel
- Concepts:** SDLC, API Integration, System Design Basics,Requirement Analysis

PROJECTS

- DPI — Deep Packet Inspection System** **GitHub:** Developed a high-performance Deep Packet Inspection (DPI) engine to analyze and filter network traffic from PCAP files using C++. Designed a robust packet processing pipeline to parse Ethernet, IP, and TCP/UDP protocols, enabling deep inspection of packet-level data.
Implemented flow-based tracking using the Five-Tuple model to manage stateful connections and ensure accurate tracking of network sessions. Integrated TLS SNI extraction to identify applications such as YouTube and Facebook from encrypted HTTPS traffic, enabling intelligent traffic classification.
Engineered a rule-based filtering system to block traffic based on IP, domain, and application type, supporting fine-grained network control. Designed and implemented a multi-threaded architecture using load balancers and fast-path worker threads to process packets in parallel.
Utilized thread-safe queues, mutexes, and condition variables to ensure safe and efficient inter-thread communication. Generated detailed traffic analytics reports including protocol distribution, application breakdown, and packet statistics.
Tech: C++, Multithreading, TCP/IP, TLS, PCAP, Data Structures, Concurrency, System Design
- FinLedger Pro — Backend Banking Infrastructure** **GitHub:** Developed a scalable backend banking system using Node.js and Express to simulate real-world financial operations, including user authentication, account management, and transaction processing. Designed a robust RESTful API architecture to handle secure and efficient client-server communication.
Implemented a double-entry ledger system to ensure transactional consistency, where every debit and credit operation is recorded atomically to maintain financial integrity. Engineered transaction workflows to handle concurrent operations and prevent data inconsistencies.
Integrated JWT-based authentication and authorization mechanisms to secure user sessions and protect sensitive financial data. Designed modular and maintainable code architecture using controllers, services, and models for better scalability and separation of concerns.
Built transaction tracking and history management features to enable auditability and traceability of financial activities. Configured email notification services for transaction alerts, enhancing system usability.
Tech: Node.js, Express.js, MongoDB, Mongoose, JWT, REST APIs, System Design

CERTIFICATIONS

- Google IT Support Professional Certificate** – Google (Google Skills)
 - The Bits and Bytes of Computer Networking** – Google (Coursera)
 - AWS Technical Essentials** – AWS Educate (Amazon)
 - Operating Systems and Systems Programming** – Cursa
 - Frontend Development Libraries** – freeCodeCamp
 - Introduction to Machine Learning** – NPTEL (IIT)
 - Market Analysis** – NPTEL(IIT)
- Jan 2026
Dec 2025
June 2025
Apr 2025
Sep 2024
Sep 2025
May 2026

POSITIONS OF RESPONSIBILITY

- Cultural Representative**
Pahadi Club, VIT Bhopal, India

 - Led cultural events, fostering teamwork, collaboration, and cross-functional coordination.
 - Strengthened project management, leadership, and communication skills through planning and executing campus initiatives.
- April 2025 – Present